TEACHING AND MENTORING STATEMENT | Naman Gupta

My teaching and mentoring philosophy strives to cultivate **intellectual rigor** and **compassionate care** among students. Akin to my research, my approach is rooted in feminist and decolonial pedagogy, promoting value sensitivities such as **empathy, self-reflection, independence, and interdependence**. I equip students to critically examine the social dimensions of modern technologies. I view teaching as a reciprocal process of learning and transformation, one that empowers students to question how systems of power shape technology and to imagine equitable futures. I believe that education can both reproduce and dismantle systems of oppression by marginalizing students through rigid academic standards. Therefore, I foster an inclusive classroom environment grounded in epistemic justice [1, 2], ensuring that students' diverse lived experiences, especially those from marginalized backgrounds, are treated as equal sources of knowledge. Such a co-learning environment encourages **vulnerability, resilience, and belonging** in a fast-paced learning environment. Moreover, the high-pressure publish-or-perish culture in R1 institutions can also be a source of mental health struggles, conflicts, and failure. I view conflict and failure as a spectrum and not as binaries, one that can be re-framed through cognitive restructuring and centering strengths and desires [3–5]. These techniques are crucial in nurturing a growth mindset by developing their **self-efficacy, agency, and drive**.

1. Teaching Process Before mentoring or teaching, I co-design a learning contract with students to cultivate trust and accountability. The learning contract helps to align expectations, provides metrics and deliverables to track their progress, feedback mechanisms and communication modalities, and proactively provides support structures to address roadblocks and conflicts. Yet, I recognize that projects and students' needs change and evolve; I am flexible in adapting to their changing needs through active listening. As an immigrant and a first-generation graduate student, I intentionally demystify the "hidden curriculum" of academia, helping students navigate institutional and professional pathways that often feel opaque. I offer resources on entire life cycle of publishing a paper: choosing an appropriate topic, why it speaks to them and why research is needed or not needed, picking the right research methods that can address their research question appropriately, collaborating as a team, submitting to the right venue, and reviewing papers through constructive criticism. Beyond papers, especially in the age of AI reliance, I hone essential skills among students to communicate their research in grant-writing, professional networking, and conversations with scholars across disciplines.

My research on support dynamics and coping mechanisms in violence [USENIX 2024] and in college contexts [CHI-LBW 2024] has enabled me to empathize with struggles that students go through in their daily lives. For example, graduate students face social isolation and mental health issues, especially after COVID-19. Moreover, Western research programs seldom center independence which may miss the collaborative nature of scientific research. Some departments do not provide adequate support structure for students to thrive. To address this, I organize a virtual peer-support community bringing together graduate students across various disciplines at UW–Madison. Finally, I designed an educational intervention [In-review CSCW 2027] that simplifies technical jargon and helped 200 GBV advocates across the US, especially in Indigenous communities. The training enabled advocates with knowledge and skills to help survivors reclaim agency. The success of these training sessions demonstrates my ability to translate computing expertise into actionable, justice-oriented education for diverse learners.

2. Teaching Experience. I practice reciprocity to impart my academic knowledge back to students. My teaching pedagogy entails careful discussion of the social implications of technology and AI in socio-cultural dynamics and how they are entangled with each other. I have been a teaching assistant (TA) at UW–Madison and IIIT–Delhi for in-person and online courses on fundamental and specialized topics. Further, I guest lectured for courses in Social Work SW623: Interpersonal Violence and Psychology and Gender & Women Studies PSYCH311/GWS311: Sexual Violence Advocacy that engaged with future social workers and students coming from diverse backgrounds.

As a TA, I have supported large and small courses on programming, data structures, and algorithms, developing instructional materials and leading discussions. As an undegrad, I TAed Mobile Computing (CSE535, $\sim$ 120 students, NPTEL, $\sim$ 430 students) and instructed refresher courses on Data Structures, Algorithms, & Systems Engineering ($\sim$ 50 graduate students). To strengthen fundamentals, I coordinated discussions, created and graded questionnaires and problem sets, invigilated laboratory exams, and offered office hours to support their learning. As a grad student, I TAed for Programming-III (CS400, $\sim$ 800 students), a course covering functional and object-oriented design patterns. I created formative and summative quizzes, assignments, and led discussion sessions and 1:1 office hours to support students. I recognized that the course is the first exposure for most sophomore students to object-oriented problem-solving, design patterns, and complex Java syntax, which contributed to their struggles. I encouraged them to deconstruct the problems into simpler steps and visually draw them out on a whiteboard, instead of blaming and shaming their coding skills. To improve learning, I contextualized how such coding patterns lead to clean and readable code for peers by sharing stories from my prior experience as a software engineer in the tech industry. These experiences taught me how to teach computer science basics while balancing practical techniques, intellectual rigor, and setting up students for their academic programs. I TAed Capstone Projects (CS639, $\sim$ 200 students) twice. I mentored senior CS and MBA students partnering with industry companies to design products to create a real-world impact, without being techno-solutionist. I guided them in design prototyping, bias mitigation, conflict resolution between teams and company mentors, and assessing the needs of the product's users. I enabled students to run a weekly SCRUM meeting, organize project tasks through sprints, and align the product with the company's short-term and long-term goals. Moreover, to make the hidden curriculum visible, I taught lectures on DevOps practices such as clean coding, version control, and agile development to cultivate skills that are missing from a typical computer science curriculum [9].

Next, I TAed for Artificial Intelligence (CS540, $\sim$ 70 students), demystifying complex components that build an AI model. I designed problem sets showing AI's relevance in CS research through applications in data mining, game design, natural language understanding, computer vision, speech recognition, and robotics. Moreover, I led weekly review classes and contextualized the social impact of AI through its uses in healthcare, recommendation algorithms, and loan disbursal systems. Such a societal contextualization is vital for early-career programmers to think critically and create real-world applications.

3. Mentoring. I actively and will continue to mentor students, organize discussion panels, social communities, and reading groups, along with advocacy efforts at Madison Tech Clinic (MTC), a community-engaged intervention that has supported 180+ survivors with their digital safety needs. I acknowledge the power differential between mentor and the student and how it informs collective learning and development. I have led informal 1:1 meetings, research meetings, and structured mentorship programs at UW–Madison and at IIIT–Delhi. To develop as a mentor, I participated in the "*Research Mentor Learning Community*" and "*Improvisation for Scientists*" to learn effective communication and mentoring pedagogy. Ten years ago, I established IIIT–Delhi's first-ever mentoring program that paired $\sim$ 200 freshman students with seniors to drive their creative ideas forward. The program was a success by providing vital exposure to students and preparing their CVs to land internships. At UW–Madison and Midwest Security Workshop, I organized three panels on navigating PhD coursework and stressors. With panelists from all research areas in CS, we guided junior students in their specific research journeys. We developed passion for research, navigating stressors, conflict, and work-life balance, and provided career advice based on the abysmal job market. Moreover, I provided 1:1 and group guidance for international students through the department mentoring program. I guided them to navigate finances, housing, and building community as they go through their transition of moving to a new country.

> *" Naman was instrumental in enabling me to begin my PhD. ...He identified and encouraged me to apply for a TA position in the Psychology Department, personally connecting me with the hiring staff. At the time, it hadn't even occurred to me to look beyond my home department, and without his guidance, I simply wouldn't have known about this opportunity. His message at that moment quite literally made it possible for me to start my PhD. Since then, he has continued to check in regularly about my TA and funding situation, and has been consistently warm and encouraging, helping me connect with social and academic communities that have made my transition to and experience of graduate life both easier and more fulfilling. "*
>
> — an international junior PhD student

4. Advocacy and Organizing. My dedication to creating collaborative research environments can be demonstrated through my passion for organizing safe spaces in reading groups and conferences. At Madison Tech Clinic, I take multiple roles to create an inclusive space. As the coordinator, I lead meetings, schedule consultations, recruit new volunteers, and design policies for clinic practice and community norms. A safe space helps consultants navigate vicarious trauma and burnout so that they can provide long-term support to each other. In the reflection meetings, the consultants share their experiences of supporting trauma survivors and co-construct knowledge by brainstorming about ever-growing vectors of tech violence. As the outreach director, I lead the outreach efforts to build partnerships with 26 GBV advocacy organizations, spreading MTC's services across 32 counties in the state of Wisconsin, US. Finally, as a consultant, I have personally supported 40+ survivors. At UW–Madison, I kick-started the Decolonial Collective that brings together interdisciplinary faculty and students from computer science, information science, history, English, education policy and psychology, curriculum & instruction, social work, and gender & women studies. The collective meetings reconcile with politically-charged topics of colonial violence and diverse thinking, which can raise conflicts and re-traumatize attendees. As the community relations lead, I ensure that attendees practice cultural humility, active listening, and support them through a written manifesto of community norms, regular check-in, and community-building exercises. These experiences have honed my facilitation skills and reaffirmed my belief in teaching as a collective process.

5. Future Courses. My experience in mentoring, teaching, and advocacy has enabled me to facilitate courses at the undergraduate and graduate levels. Further, I plan to teach fundamental courses and further develop special topics courses such as "Decolonial Race, Gender, and Technology".

- Fundamentals of Computer Science: Introduction to Programming, Python, and Data Science
- Methods: Qualitative Research Methods, Community-Engaged Research, Decolonial Praxis and Research Methods
- Human-Computer Interaction: Introduction to HCI, HCI in Indigenous and Global South, Decolonial HCI, Feminist HCI
- Security and Privacy: Computer Security and Privacy, Human-Centered Security and Privacy
- Information: Gender (or Race) and Information Technology, Indigenous Epistemology and Technology, STS
- Ethics: Ethics, Society, Justice in Computing, Critical Computing, History of Science and Technology, Value Sensitive, Responsible, Inclusive Design, Ethics of AI, Ethics of Data Science, Ethics of Emerging Technologies
- Health: Healthcare and Technology, Digital Safety for Marginalized Communities

My Publications

[USENIX 2024] **Naman Gupta**, Kate Walsh, Sanchari Das, and Rahul Chatterjee. ""I really just leaned on my community for support": Barriers, Challenges, and Coping Mechanisms Used by Survivors of {Technology-Facilitated} Abuse to Seek Social Support". In: 2024. URL: https://dl.acm.org/doi/10.5555/3698900.3699179.

[CHI-LBW 2024] **Naman Gupta**, Sanchari Das, Kate Walsh, and Rahul Chatterjee. "A Critical Analysis of the Prevalence of Technology-Facilitated Abuse in US College Students". In: *Extended Abstracts of the 2024 CHI Conference on Human Factors in Computing Systems*. 2024. URL: https://doi.org/10.1145/3613905.3652036.

[In-review CSCW 2027] **Naman Gupta**, Connie W. Chau, Sophie Stephenson, Rahul Chatterjee, and Kate Walsh. "Lessons from Trauma-Informed Training on Technology-Facilitated Abuse for Gender-Based Violence Advocates". In: *review at CSCW* (2026). URL: https://arxiv.org/abs/2610.02688.

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